

Yuhan Ma

emmmmyh.inori@gmail.com | +1 858-247-8561

PROFESSIONAL SUMMARY

LLM Post-Training / Agentic RL Engineer with 2+ years of experience in AI research and model training, specializing in domain adaptation, reinforcement learning, and high-performance inference. Proficient in Python, PyTorch, C++, full-stack development, and distributed training and serving.

EDUCATION

University of California San Diego | M.S. in Computer Science | September 2026 – June 2028 (Expected)

Tsinghua University | B.Eng. | September 2018 – July 2023

EXPERIENCE

AITOPIA Artificial Intelligence Technology Co., Ltd. | Machine Learning Engineer | July 2023 – April 2026

Research Report Factor Extraction and Evolution Agent

Python · Qwen3-4B · verl · GRPO · vLLM · FSDP · Pydantic · FastAPI · Qlib · W&B

- Defined a 7-field **FactorSpec** schema and built a dataset of 3,000 annotated factor examples from quantitative research report, enforcing institution/series-disjoint training and held-out splits.
- Built a multi-turn report-reading environment with tools for search, pagination, and schema validation; converted annotations into **SFT** trajectories with loss restricted to assistant tool calls and final responses.
- Implemented a programmatic **verifier** for JSON validity, expression executability, and source grounding, gating entry into factor evolution and testing reward robustness against five adversarial error categories.
- Built a factor-evolution loop supporting up to four backtest evaluations per trajectory, allowing expression and lookback-window edits while penalizing changes to universe, frequency, and neutralization settings.
- Post-trained separate extraction and evolution **LoRA** adapters on **Qwen3-4B** via **SFT** followed by **GRPO**, using **FSDP** for distributed training, **vLLM** for rollout, and **verl AgentLoop** for multi-turn interaction; used verification-based and backtest-based rewards for extraction and evolution, respectively.
- Improved end-to-end FactorSpec success from 56% to 68% and evidence grounding from 60% to 82% through **GRPO**.

Financial LLM Post-Training and Inference Service

Python · Llama-2-13B · Transformers · PEFT · FastAPI · vLLM · Prometheus

- Curated approximately 50K seed examples across six financial NLP tasks, recording data source and creating source-disjoint training and held-out splits.
- Fine-tuned **Llama-2-13B** with **LoRA** and **task-aware sampling**, tracking experiments in **Weights & Biases**; improved classification **Macro-F1** from 63.8 to 72.4 and numerical-QA accuracy from 68.5% to 77.1%.
- Established 10 adversarial evaluation suites covering 5,000 cases, linking each release to its base model, adapter configuration, and evaluation report for traceability.
- Implemented a **FastAPI** inference gateway with authentication, tenant quotas, model registration, runtime LoRA loading, and adapter-version rollback.
- Deployed the gateway on **Kubernetes** in front of **vLLM**, using vLLM continuous batching for online inference.

ETF Create/Redeem Pricing and Limit-Order Execution

Python · C++ · Poisson/Hawkes Processes · Reinforcement Learning · Decision Making

- Developed pricing and order-execution components for a live ETF strategy managing approximately RMB 500 million in assets; the overall strategy achieved approximately 15% annualized excess return over the evaluation period.
- Validated pricing features through walk-forward evaluation and used model signals to guide trade direction, participation rates, and execution price bands.
- Modeled state-dependent order-arrival intensities with **Hawkes processes**, deriving fill probabilities and queue-depletion rates from clustered activity.
- Built an event-driven **LOB** replay environment with price-time priority and partial-fill handling, resampling Hawkes flows for training diversity.
- Formulated execution as an **MDP** over queue position, remaining quantity, and NAV residual, with rewards balancing shortfall against inventory.
- Trained a **PPO** policy to select limit-price levels, cancel or replace orders, and execute partial or full market orders.
- Evaluated the policy against TWAP and VWAP on held-out data, improving fill rate from 2.0% to 2.3% and reducing implementation shortfall by 0.3 bp.

RESEARCH

LUCiD: Mitigating Hallucinations in LLMs with Learned Unilayer Contrastive Decoding

Manuscript under double-blind review | Implementation and Experimental Evaluation

- Co-developed a contrastive-decoding method that freezes the **LLM** backbone while training a single middle-layer residual probe and per-token gate.
- Trained the probe under an adaptive plausibility constraint, keeping the base model's weights entirely untouched throughout the decoding process.
- Ran depth sweeps and component ablations across five models spanning 0.6B to 12B parameters, aggregating results over ten held-out benchmark sets.
- Outperformed **Vanilla**, **DoLa**, and **SLED** baselines on **SQuAD** for four of five tested models, validating the approach consistently across model scales.
- Lifted **LLaMA-3-8B** SQuAD F1 from 0.696 to 0.808 under sampling decoding, holding peak-memory overhead to just 1.2% or less versus the base model.

COMPETITIONS AND HONORS

- Jane Street Real-Time Market Data Forecasting, Kaggle Silver Medal** | Top 3% of 3,757 teams (July 2025) | blending **LightGBM**, **XGBoost**, and **CatBoost** models; Stacked **Ridge**, **ElasticNet**, and **MLP** models, using Bayesian-optimized hyperparameter.
- Xilinx All Women Hackathon (Oct 2021), Outstanding Award** | built a Xilinx PYNQ autonomous line-following racing car with sensor fusion and PWM motor control.

SKILLS

Programming: Python, C++, SQL, Java, Pandas, NumPy, HTML/CSS

LLM training, Inference & Serving: PyTorch, Transformers, SFT, GRPO, LoRA/PEFT, verl, TRL, FSDP, W&B, ONNX, vLLM, SGLang

Full-Stack: FastAPI, Node.js, TypeScript, React, RESTful APIs, PostgreSQL, Redis, Docker, Kubernetes, HTML/CSS, Linux, Git, CI/CD, AWS, Kafka

Statistics Modeling: Probability and Statistics, Stochastic Processes, Time-series mModeling, Bayesian Optimization